

Confidence-Guided Markov Weighting for Semi-Supervised Tabla Stroke Transcription

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Abstract—Tabla Stroke Transcription (TST) converts tabla audio into symbolic stroke sequences, but the scarcity of annotated recordings makes fully supervised training challenging. We propose a semi-supervised framework that uses sequence-level labelled and unlabelled tabla recordings. A teacher model generates pseudo-label sequences, while a Stroke-Level Confidence Estimation Model (S-CEM) estimates confidence for each predicted stroke. To improve training with uncertain pseudo-labels, we propose Confidence-Guided Markov Weighted Alternative Temporal Classification (CMW-ATC), which weights candidate sequences within uncertain spans using learned stroke transitions and reliable neighbouring strokes. Experiments across three TST evaluation settings show consistent improvements over conventional teacher-student training and Alternative Temporal Classification in its replacement form (ATC-R). Ablation studies further show the contributions of S-CEM, Markov transition weighting, and the use of reliable strokes on both sides of uncertain spans.

Index Terms—Confidence estimation, CTC, Markov weighting, semi-supervised learning, tabla transcription

I. INTRODUCTION

Tabla Stroke Transcription (TST) converts tabla audio into symbolic stroke sequences, supporting applications such as *tāla* identification, rhythmic-pattern analysis, pedagogy, and digital preservation. Developing accurate TST systems remains challenging because of the scarcity of annotated data. Many supervised approaches require precise onset-level annotations, which are expensive and time-consuming to obtain and typically require expert tabla players.

Previous TST studies have explored onset-based stroke segmentation and classification [1], [2], as well as strongly supervised deep-learning approaches [3], [4]. Meta-learning has further enabled TST with limited onset-level annotated data [5]. The Adaptive Dynamic Rhythm Language Model (ADRM) enabled TST from sequence-level stroke annotations without explicit onset timings [6]. Although this reduces the annotation detail required for TST, sequence-level stroke annotations are still required for training.

Unsupervised learning can further reduce dependence on manual annotation. Recent work has used acoustic

clustering to learn stroke representations across tabla and other Indian percussion instruments without predefined stroke labels [7]. Beyond purely unsupervised approaches, unlabelled recordings can also be incorporated through semi-supervised learning, where the available sequence-level annotations continue to provide supervision. In this setting, a teacher model generates pseudo-labels for the unlabelled recordings, but errors in these sequences can propagate to the student model.

Connectionist Temporal Classification (CTC) [8] enables training from sequence-level labels without explicit frame-level alignment, but assumes that the supplied target sequence is correct. Graph-based extensions have relaxed this assumption by allowing alternative supervision structures [9]–[13]. Alternative Temporal Classification (ATC) [14] further uses pseudo-label confidence to relax uncertain positions during semi-supervised training. In its replacement form, ATC-R assigns the same alternative weight to all non-blank labels at an uncertain position and does not exploit the sequential relationships among stroke labels.

This limitation is particularly relevant in situations where multiple consecutive pseudo-labels are uncertain, since an uncertain stroke can provide unreliable context for the next position. A conventional Markov model uses forward transitions conditioned on the preceding stroke, whereas a Markov bridge conditions the intermediate path on both endpoint states [15], [16]. To the best of our knowledge, this principle has not been used for uncertainty-aware pseudo-label supervision in speech or music transcription. Motivated by this principle, we propose Confidence-Guided Markov Weighted Alternative Temporal Classification (CMW-ATC), which uses learned stroke transitions and the available reliable boundary information to weight candidate sequences within uncertain spans.

Identifying which pseudo-label strokes should be treated as uncertain also requires reliable stroke-level confidence estimates. Existing confidence estimation methods have primarily been developed for words or subwords in ASR [17], whereas tabla strokes are atomic prediction classes. We therefore develop a Stroke-Level Confidence Estimation Model (S-CEM) that predicts the confidence that each teacher-generated stroke is correct from teacher-derived representations under sequence-

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level supervision. The main contributions of this work are:

- 1) We introduce CMW-ATC for uncertainty-aware semi-supervised pseudo-label training. It forms candidate stroke sequences within uncertain spans and assigns them weights using learned stroke transitions and available reliable boundaries.
- 2) We introduce S-CEM for stroke-level confidence estimation. It estimates the correctness of each teacher-predicted stroke using acoustic representations and CTC posterior information.
- 3) We conduct component ablation studies across three TST evaluation settings. The experiments separately examine the contributions of S-CEM, forward Markov weighting, and right-boundary conditioning.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed methodology. Section IV describes the experimental setup, followed by results and discussion in Section V. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Tabla Stroke Transcription

Research on Tabla Stroke Transcription (TST) has explored acoustic stroke characterization and automatic recognition from tabla recordings. Early approaches followed a segment-and-classify paradigm using acoustic features and statistical models [1], [18]. Subsequent studies investigated the acoustic characteristics of tabla strokes [19], temporal relationships between successive strokes [20], and onset detection for identifying stroke boundaries [2]. Related work on mridangam has also explored acoustic and statistical approaches to stroke transcription [21]–[23].

More recent TST approaches have adopted supervised deep-learning models, including models adapted from automatic drum transcription [3] and transfer-learning approaches using Western drum data [4]. These approaches rely on onset-level stroke annotations. Meta-learning has further improved data efficiency by enabling TST with a limited amount of onset-annotated data [5], but still requires temporal annotations. Our previous work introduced the Adaptive Dynamic Rhythm Language Model (ADRM), which enabled sequence-level TST without explicit onset timings and incorporated rhythmic information during CTC-lattice decoding [6]. However, ADRM still requires sequence-level stroke annotations for training.

Unsupervised learning provides another direction for reducing dependence on manual annotation. Recent work has used timbre-based clustering to learn stroke representations across tabla and mridangam without predefined

stroke labels [7]. The resulting clusters can subsequently be mapped to the target stroke vocabulary using labelled data.

B. Graph-Based Extensions of CTC

Connectionist Temporal Classification (CTC) [8] provides an alignment-free objective for sequence-level supervision and is therefore well suited to TST when only ordered stroke sequences are available. However, standard CTC assumes that the supplied target sequence is correct. Several extensions modify the CTC training graph to handle incomplete or erroneous supervision. W-CTC [10] introduces wildcard transitions for partial label sequences, while STC [9] allows missing labels at arbitrary positions. BTC [11] introduces bypass transitions for imperfect transcripts, and OTC [12] combines wildcard and bypass mechanisms to accommodate different transcription errors.

Graph-based Temporal Classification (GTC) [13] generalizes CTC to weighted graph supervision, allowing multiple candidate target sequences and their weights to be marginalized during training. In semi-supervised ASR, GTC constructs supervision graphs from multiple pseudo-label hypotheses and their associated scores, providing a general framework for weighted alternative supervision.

Markov models provide a standard framework for modeling state transitions, whereas a Markov bridge describes a Markov path conditioned on specified endpoint states [15]. Such endpoint-conditioned paths have also been used for statistical inference in Markov processes [16]. This formulation is relevant to uncertain sequence spans because reliable states before and after an uncertain region can provide additional information for weighting the possible intermediate sequences.

Alternative Temporal Classification (ATC) [14] uses pseudo-label confidence to identify uncertain tokens during semi-supervised training. Its ATC-R variant replaces a low-confidence token with a wildcard that accepts any non-blank label, but does not use the sequential relationships among these alternatives when assigning their weights.

These lines of work provide the basis for our approach. ATC provides confidence-guided alternative supervision, while Markov-bridge conditioning provides a way to use reliable neighbouring states when weighting uncertain intermediate sequences. CMW-ATC combines these principles by using learned stroke transitions and the available reliable boundary information to weight candidate sequences within uncertain pseudo-label spans.

C. Confidence Estimation

Confidence estimation is widely used to assess the reliability of model predictions. Posterior probability and entropy provide simple confidence measures, but neural models can produce overconfident predictions [24], [25]. Auxiliary Confidence Estimation Models (CEMs) instead learn confidence from model-derived representations. In ASR, CEMs have been trained using binary correctness targets obtained by aligning hypotheses with reference sequences [26]. TeLeS [27] introduced continuous targets that combine temporal and lexical similarity, but requires temporal alignment between hypothesis and reference. TruCLEs [17] avoids this temporal-alignment requirement by constructing continuous targets from true-class probabilities and lexical similarity.

These approaches were developed for ASR, where outputs are words or subwords and lexical similarity can represent partial correctness. In TST, each tabla stroke is an atomic prediction class, so lexical similarity between stroke labels does not provide a meaningful measure of transcription correctness. We therefore use a task-specific stroke-level CEM with binary correctness targets under sequence-level supervision.

III. PROPOSED METHOD

The proposed framework consists of teacher-generated pseudo-labels, stroke-level confidence estimation, and confidence-guided student training. The student is trained using conventional Connectionist Temporal Classification (CTC) on labelled recordings and the proposed Confidence-Guided Markov-Weighted Alternative Temporal Classification (CMW-ATC) objective on pseudo-labelled recordings. CMW-ATC uses stroke-level confidence and learned stroke transitions to guide training on pseudo-labelled recordings.

Let $X \in \mathbb{R}^{F \times T}$ denote an input acoustic feature matrix consisting of F feature coefficients over T frames, and let $Y = [y_1, \dots, y_U] \in \mathcal{S}^U$ denote a corresponding stroke sequence of length U , where $y_u \in \mathcal{S}$ and $\mathcal{S} = \{s_1, s_2, \dots, s_K\}$ is the tabla stroke vocabulary. The labelled and unlabelled datasets are denoted by $D_L = \{(X_i, Y_i)\}_{i=1}^N$ and $D_U = \{X_j\}_{j=1}^M$, respectively, where N and M denote the numbers of labelled and unlabelled recordings.

A. CTC Preliminaries

Connectionist Temporal Classification (CTC) enables sequence prediction without requiring explicit frame-level alignment between the input audio and target stroke sequence. It introduces a blank symbol and marginalizes over all frame-level paths that collapse to the target sequence. Let $\bar{\mathcal{S}} = \mathcal{S} \cup \{\emptyset\}$ denote the CTC output

label set, where $\emptyset$ is the blank symbol, and let $\mathcal{B}$ denote the CTC collapse function that merges consecutive repetitions and removes blanks. For input X and target stroke sequence $Y = [y_1, \dots, y_U]$, the CTC probability is

$$P_{\theta}^{\text{CTC}}(Y | X) = \sum_{\pi: \mathcal{B}(\pi)=Y} \prod_{t=1}^T p_{\theta}(\pi_t | X, t), \quad (1)$$

where $\pi \in \bar{\mathcal{S}}^T$ is a frame-level CTC path. The corresponding training loss is

$$\mathcal{L}_{\text{CTC}}(X, Y) = -\log P_{\theta}^{\text{CTC}}(Y | X). \quad (2)$$

This CTC formulation is used for labelled recordings, while CMW-ATC extends it to pseudo-labelled recordings by marginalizing over alternative stroke sequences at uncertain positions.

B. Teacher Model and Pseudo-Label Generation

A teacher acoustic model with parameters θ_T is trained on the labelled dataset using the CTC loss in Eq. (2). After training, the teacher is frozen and used to generate pseudo-label sequences for the unlabelled recordings. For an input X , the teacher produces a pseudo-label sequence $\hat{Y} = [\hat{y}_1, \dots, \hat{y}_{\hat{U}}]$ using CTC decoding, where $\hat{U}$ is the length of the decoded sequence. The stroke-transition model is not used during teacher decoding, so pseudo-labels are generated only by the acoustic model.

C. Stroke-Level Confidence Estimation

Pseudo-labelled sequences may contain stroke predictions with different levels of reliability. We therefore adapt the Confidence Estimation Model (CEM) framework of TruCLEs [17] to develop a Stroke-Level Confidence Estimation Model (S-CEM) for TST. The S-CEM uses teacher representations for each predicted stroke to estimate stroke-level confidence.

Unlike ASR words, tabla strokes are discrete stroke labels. Consequently, lexical similarity between stroke labels does not provide a meaningful measure of partial correctness. We therefore replace the continuous target used by TruCLEs, which includes lexical similarity, with a binary target indicating whether each predicted stroke is correct.

For S-CEM training, the same frozen teacher is applied to the labelled training recordings to obtain predicted stroke sequences and teacher representations. For an input recording X , let $\hat{Y} = [\hat{y}_1, \dots, \hat{y}_{\hat{U}}]$ denote the teacher prediction and $Y = [y_1, \dots, y_U]$ the reference stroke sequence. The two sequences are aligned using stroke-level Levenshtein alignment. For each predicted stroke $\hat{y}_u$, the binary correctness target is defined as

$$b_u = \begin{cases} 1, & \text{if } \hat{y}_u \text{ is matched to the same reference stroke,} \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

A deletion does not produce a predicted stroke and therefore has no corresponding stroke-level target. The Levenshtein alignment is used only to determine the correctness targets and does not require onset-level annotations.

To obtain a teacher representation for each predicted stroke occurrence, we compute the most probable CTC alignment. The constrained Viterbi alignment is

$$\pi^* = \arg \max_{\substack{\pi \in \mathcal{S}^T \\ \mathcal{B}(\pi) = \hat{Y}}} P_{\theta_T}(\pi | X), \quad (4)$$

where θ_T denotes the frozen teacher parameters. Each predicted stroke occurrence is treated separately in the alignment, so repeated occurrences of the same stroke label remain distinct.

Let $\mathcal{T}_u$ denote the non-blank output indices assigned by π^* to the u th predicted stroke occurrence. Let $r_t \in \mathbb{R}^D$ denote the teacher representation immediately before the CTC output layer, and let $p_t \in \mathbb{R}^{K+1}$ denote the corresponding CTC posterior vector, where the additional dimension corresponds to the CTC blank. The stroke-level representations are obtained by temporal averaging:

$$\bar{r}_u = \frac{1}{|\mathcal{T}_u|} \sum_{t \in \mathcal{T}_u} r_t, \quad \bar{p}_u = \frac{1}{|\mathcal{T}_u|} \sum_{t \in \mathcal{T}_u} p_t. \quad (5)$$

The resulting stroke-level feature vector is

$$\phi_u = [\bar{r}_u; \bar{p}_u] \in \mathbb{R}^{D+K+1}, \quad (6)$$

which is used as input to the S-CEM. The S-CEM produces a continuous stroke-level confidence score

$$c_u = S\text{-CEM}_{\theta_c}(\phi_u), \quad c_u \in (0, 1). \quad (7)$$

The S-CEM is trained using binary cross-entropy:

$$\mathcal{L}_{\text{S-CEM}} = - \sum_{u=1}^{\hat{U}} [b_u \log c_u + (1 - b_u) \log(1 - c_u)] \quad (8)$$

Higher c_u indicates greater confidence that the predicted stroke is correct.

For an unlabelled recording, the teacher prediction and constrained CTC alignment provide the stroke-level feature vectors ϕ_u , from which the trained S-CEM produces a confidence score c_u for each predicted stroke. These scores are used to identify uncertain pseudo-label positions during CMW-ATC student training.

D. Confidence-Guided Markov-Weighted Alternative Temporal Classification

CMW-ATC uses the stroke-level confidence score to determine how each pseudo-label position is treated during training. If $c_u \geq \tau$, the corresponding pseudo-label is treated as reliable and kept fixed, although it is not regarded as ground truth. If $c_u < \tau$, the position is treated as uncertain and all strokes in $\mathcal{S}$ are allowed as alternatives. Similar to the replacement form of Alternative Temporal Classification (ATC-R) [14], CMW-ATC allows alternative labels at uncertain positions. However, unlike ATC-R, CMW-ATC uses reliable neighbouring pseudo-labels and learned stroke-transition probabilities to weight these alternative sequences differently.

1) *Stroke Transition Model*: We estimate a first-order Markov model from the original labelled training sequences. If $N(s_i, s_j)$ is the number of times s_j follows s_i , add-one smoothing gives

$$P(s_j | s_i) = \frac{N(s_i, s_j) + 1}{\sum_{s' \in \mathcal{S}} N(s_i, s') + K}. \quad (9)$$

Pseudo-labels and validation or test sequences are not used. The resulting transition matrix P remains fixed during student training. We also estimate the initial-state distribution from training data using add-one smoothing:

$$\rho(s_j) = \frac{N_{\text{first}}(s_j) + 1}{\sum_{k=1}^K N_{\text{first}}(s_k) + K}, \quad (10)$$

where $N_{\text{first}}(s_j)$ counts labelled sequences whose first stroke is s_j .

2) *Markov Weighting of Uncertain Stroke Sequences*: Consecutive low-confidence positions are grouped into maximal uncertain spans. Let $[a, b]$ denote such a span and let $m = b - a + 1$ denote its length. Thus, $c_u < \tau$ for all $u \in [a, b]$, while the neighbouring positions, when present, are reliable. For a single-position span, the corresponding empty product is taken as one and the empty sum as zero.

The first-order Markov model defined in Section III-D1 assigns the probability of a sequence of m strokes $z_{1:m} = [z_1, \dots, z_m]$ as

$$P(z_{1:m}) = \rho(z_1) \prod_{i=2}^m P(z_i | z_{i-1}), \quad (11)$$

where ρ is the initial-state distribution and $P(z_i | z_{i-1})$ is the learned stroke-transition probability. However, pseudo-label sequences generated by the teacher are not necessarily reliable at every position. When low-confidence positions occur consecutively, their predicted strokes may not be correct.

Candidate sequences within an uncertain span are weighted using the stroke-transition probabilities and initial-state distribution defined in Section III-D1. The weighting of an uncertain span depends on the available reliable neighbouring pseudo-labels. A reliable boundary may be available on both sides, only on the left, only on the right, or on neither side. These four cases determine how the candidate sequences are weighted. When reliable pseudo-labels are available on both sides of the span ($a > 1$ and $b < \hat{U}$), let $\ell = \hat{y}_{a-1}$ and $r = \hat{y}_{b+1}$ denote the reliable left and right boundary strokes, respectively.

For a candidate sequence $z_{a:b} = [z_a, \dots, z_b]$, the probability of the path from ℓ through the candidate sequence to r is

$$P(z_{a:b}, r | \ell) = P(z_a | \ell) \prod_{i=a+1}^b P(z_i | z_{i-1}) P(r | z_b). \quad (12)$$

Since this path contains $m + 1$ transitions from ℓ to r , summing its probability over all possible length- m candidate sequences gives $(P^{m+1})_{\ell r}$. Conditioning on the two reliable boundaries therefore gives the Markov bridge distribution

$$q_{\text{BB}}(z_{a:b} | \ell, r) = \frac{P(z_a | \ell) \prod_{i=a+1}^b P(z_i | z_{i-1}) P(r | z_b)}{(P^{m+1})_{\ell r}}. \quad (13)$$

If the uncertain span reaches the end of the pseudo-label sequence ($a > 1$ and $b = \hat{U}$), the reliable left boundary $\ell = \hat{y}_{a-1}$ remains available, but there is no right boundary. The candidate distribution is therefore

$$q_L(z_{a:b} | \ell) = P(z_a | \ell) \prod_{i=a+1}^b P(z_i | z_{i-1}). \quad (14)$$

Because the transition matrix is row-normalized, summing this expression over all length- m candidate sequences gives one, so no additional normalization term is required.

Conversely, if the uncertain span begins at the first pseudo-label position ($a = 1$ and $b < \hat{U}$), no reliable left boundary is available. The first candidate stroke is weighted by the initial-state distribution, while the reliable right boundary $r = \hat{y}_{b+1}$ is included in the path probability:

$$q_R(z_{a:b} | r) = \frac{\rho(z_a) \prod_{i=a+1}^b P(z_i | z_{i-1}) P(r | z_b)}{(\rho P^m)_r}. \quad (15)$$

The denominator $(\rho P^m)_r$ is the total probability of reaching r after the m candidate positions and normalizes the distribution over all possible candidate sequences.

Finally, if the uncertain span covers the entire pseudo-label sequence ($a = 1$ and $b = \hat{U}$), neither boundary is available. The candidate distribution then reduces to the initial-state Markov distribution:

$$q_{\emptyset}(z_{1:m}) = \rho(z_1) \prod_{i=2}^m P(z_i | z_{i-1}). \quad (16)$$

Since both ρ and the transition matrix are normalized, $q_{\emptyset}$ also sums to one over all length- m candidate sequences. Thus, q_{BB} , q_L , q_R , and $q_{\emptyset}$ covers all possible boundary cases of an uncertain span, while using reliable neighbouring pseudo-labels when available.

3) *Candidate Graph Construction*: For each pseudo-label position u , define the candidate-label set

$$\mathcal{C}_u = \begin{cases} \{\hat{y}_u\}, & c_u \geq \tau, \\ \mathcal{S}, & c_u < \tau. \end{cases} \quad (17)$$

A candidate sequence is denoted by $Z = [z_1, \dots, z_{\hat{U}}]$. The candidate space is therefore

$$\mathcal{Z} = \mathcal{C}_1 \times \mathcal{C}_2 \times \dots \times \mathcal{C}_{\hat{U}}. \quad (18)$$

The four boundary cases derived in Section III-D2 can be represented in a single graph form. For an uncertain span $[a, b]$, the left and right boundary factors $g_L(z_a)$ and $g_R(z_b)$ are defined based on whether reliable boundary strokes are available:

$$\begin{aligned} g_L(z_a) &= \begin{cases} P(z_a | \ell), & \ell \text{ available,} \\ \rho(z_a), & \ell \text{ unavailable,} \end{cases} \\ g_R(z_b) &= \begin{cases} P(r | z_b), & r \text{ available,} \\ 1, & r \text{ unavailable.} \end{cases} \end{aligned} \quad (19)$$

where $\ell = \hat{y}_{a-1}$ and $r = \hat{y}_{b+1}$ denote the reliable strokes before and after the span, respectively, when available. The Markov log-score of a candidate sequence is then defined as

$$\begin{aligned} A_{\text{CMW}}(z_{a:b}) &= \log g_L(z_a) + \sum_{i=a+1}^b \log P(z_i | z_{i-1}) \\ &\quad + \log g_R(z_b). \end{aligned} \quad (20)$$

This score combines the left boundary factor, the transitions between candidate strokes within the span, and the right boundary factor. The candidate graph preserves the length of the teacher pseudo-label sequence and allows alternative stroke labels only at uncertain positions.

4) *CMW-ATC Training Objective*: For an uncertain span $[a, b]$, the Markov scores from Eq. (20) are normalized over all candidate sequences to obtain their weights:

$$q(z_{a:b}) = \frac{\exp A_{\text{CMW}}(z_{a:b})}{\sum_{\tilde{z}_{a:b} \in \mathcal{Z}_{a:b}} \exp A_{\text{CMW}}(\tilde{z}_{a:b})}, \quad (21)$$

where $\mathcal{Z}_{a:b} = \mathcal{C}_a \times \dots \times \mathcal{C}_b$ denotes the candidate space for the span. Depending on the boundary case, this weight corresponds to q_{BB} , q_L , q_R , or $q_\emptyset$ defined in Section III-D2. If a sequence contains J separated uncertain spans, the weight of a candidate sequence is

$$q_{\text{CMW}}(Z) = \prod_{j=1}^J q(z_{a_j:b_j}), \quad (22)$$

where $[a_j, b_j]$ denotes the j -th uncertain span. When $J = 0$, the only candidate sequence is the teacher prediction, and its weight is one. Since each uncertain-span distribution is normalized, the sequence weights satisfy $\sum_{Z \in \mathcal{Z}} q_{\text{CMW}}(Z) = 1$.

The student model assigns a CTC probability to each candidate sequence. The loss for unlabelled data is therefore

$$\mathcal{L}_{\text{CMW-ATC}}(X, \hat{Y}) = -\log \sum_{Z \in \mathcal{Z}} q_{\text{CMW}}(Z) P_{\theta_s}^{\text{CTC}}(Z | X). \quad (23)$$

The weighted sum over candidate sequences is computed on the candidate graph without explicitly enumerating all candidate sequences. The overall student loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CTC}}(X_L, Y_L) + \lambda \mathcal{L}_{\text{CMW-ATC}}(X_U, \hat{Y}_U), \quad (24)$$

where λ controls the contribution of the unlabelled data.

IV. EXPERIMENTAL SETUP

This section describes the datasets, model configuration, training procedure, baselines, ablation studies, and hyperparameter settings used for evaluation.

A. Datasets

We use four source datasets for Tabla Stroke Transcription (TST): Tabla Identification Dataset (TID), Tabla Solo Dataset (TSD), Hindustani Music Rhythm Dataset (HMR), and TID-S. TID and TID-S were introduced in [6], while TSD [20] and HMR [28] are publicly available. These are organized into three evaluation sets: *TID+TSD*, *HMR*, and *TID-S*. TSD is combined with TID because it contains limited labelled training data. The three settings cover tabla solo recordings, Hindustani music with vocal and instrumental accompaniment, and synthetic tabla recordings with controlled rhythmic variation. Their training, validation, and test partitions are summarized in Table I, with further dataset details provided in Supplementary Material (S.I).

The labelled training partitions are augmented following [3] using pitch shifting, time scaling, attack remixing, spectral filtering, and stroke remixing; validation and test data are not augmented. For semi-supervised learning, we additionally use an unlabelled YouTube corpus containing 676 minutes of tabla solo performances and 602

minutes of Hindustani concert recordings. The same unlabelled corpus is used for all semi-supervised methods.

B. Feature Extraction and Model Architecture

All experiments use 128-dimensional log-Mel spectrograms with a 2048-sample analysis window and a 10 ms frame shift at 44.1 kHz. The features are mean-normalized per frequency. The teacher and student use the C-TDNN-F architecture adopted for TST in [6], based on factorized TDNNs [29] and convolutional TDNN-F models [30]. The model consists of 12 C-TDNN-F layers with bottleneck projections, temporal convolution with context $[-1, 0, +1]$, residual connections, batch normalization, and dropout (0.1), followed by a linear projection and CTC output layer. Increasing dilation factors are used to expand the temporal receptive field. The same architecture is used for all learning-based semi-supervised methods.

1) *Confidence Estimation Model*: S-CEM adapts the CEM architecture used in the CTC-based TruCLEs framework [17] to stroke-level confidence estimation, with three fully connected layers of 512, 256, and 128 units, ReLU hidden activations, and a sigmoid output. It is trained using Adam with a learning rate of 10^{-3} for 100 epochs. S-CEM is kept fixed during student training.

C. Semi-Supervised Training Procedure

A separate CTC teacher is trained for each evaluation setting using its corresponding labelled training data. After training, each teacher is frozen and used to obtain the predictions and representations required for S-CEM training and to generate pseudo-labels for the unlabelled recordings. For each evaluation setting, P and ρ are estimated independently from its original labelled training sequences and remain fixed during student training. Augmented sequences, pseudo-labels, validation data, and test data are not used for their estimation. A separate student is trained from scratch for each evaluation setting using conventional CTC on labelled recordings and CMW-ATC on pseudo-labelled recordings.

D. Baseline Methods

All methods are evaluated as symbolic stroke sequences using the same sequence-level Stroke Error Rate (SER). We consider the onset-level supervised OTM [4] and Segment-and-Classify (S&C) [1] methods, together with the stroke-clustering framework in [7]. OTM and S&C are evaluated on HMR, while clustering is evaluated on all three evaluation settings. S&C retains its onset-based segmentation and is adapted to the 30-stroke vocabulary. The clustering method does not use stroke

labels during cluster formation; labelled training data are used only to map the resulting clusters to the target stroke vocabulary.

For semi-supervised learning, we consider a conventional CTC teacher–student baseline and ATC-R [14]. The teacher–student baseline trains on teacher-generated pseudo-labels using conventional CTC. ATC-R retains its Contrastive-CTC training, CTC-based token confidence, automatic thresholding, and scaled wildcard replacement. Unlike the original ATC-R framework, the teacher is kept fixed during student training to match the frozen-teacher protocol used by CMW-ATC. Additional baseline details are provided in Supplementary Sections S.II and S.III.

E. Ablation Studies

The ablation study examines the two main components of the proposed framework: stroke-level confidence estimation and Markov-based candidate weighting. ATC-R [14] is included as the reference alternative-label method. A1 combines S-CEM confidence with the scaled wildcard replacement of ATC-R. A2 uses S-CEM with all-stroke candidates and assigns equal weight to all candidate sequences, without Markov weighting. A3 retains the same confidence and candidates as A2, but applies forward Markov weighting using the reliable left boundary, when available, and the within-span transitions, without using the right boundary.

The proposed CMW-ATC additionally uses the reliable right boundary when available. Thus, A2 versus A3 evaluates the contribution of Markov transition weighting, while A3 versus CMW-ATC evaluates the contribution of the right reliable boundary. To separately evaluate confidence estimation, A4 uses CTC-based token confidence with the same CMW weighting. Therefore, A4 versus the proposed system evaluates the contribution of S-CEM while keeping the candidate formulation and Markov weighting unchanged.

F. Hyperparameter Settings

CMW-ATC has two task-specific hyperparameters: the confidence threshold τ and the pseudo-label loss weight λ . The transition matrix P and initial-state distribution ρ use fixed add-one smoothing and have no tunable hyperparameters. The teacher and student models are trained using Adam with a learning rate of 10^{-3} for 100 epochs. The confidence threshold and loss weight are selected jointly on the corresponding validation set from $\tau \in \{0.4, 0.5, \dots, 0.9\}$ and $\lambda \in \{0.1, 0.2, \dots, 1.0\}$. The selected values are $(\tau, \lambda) = (0.6, 0.5)$ for TID+TSD and TID-S, and $(\tau, \lambda) = (0.5, 0.4)$ for HMR.

For ATC-R, the wildcard scale is tuned on the validation set to $\eta = 0.4$ and then held fixed for test evaluation;

its original automatic confidence-thresholding procedure is retained. For A4, the CTC-based confidence threshold is selected on the corresponding validation set from the same threshold range used for S-CEM. The remaining ablations use the corresponding CMW-ATC hyperparameters. Decoding beam-search parameters are $k_{\text{beam}} = 120$ and $\Delta_{\text{beam}} = 10$, while CMW candidate weighting is computed on the full candidate graph without beam pruning.

V. RESULTS AND DISCUSSION

Performance is evaluated using Stroke Error Rate (SER), defined as

$$\text{SER}(\%) = \frac{S + D + I}{N} \times 100, \quad (25)$$

where S , D , and I denote the numbers of substitution, deletion, and insertion errors, respectively, and N is the number of reference strokes. Lower SER indicates better transcription performance. Unless otherwise stated, all results are reported on the corresponding test sets.

Table II presents the transcription results across the evaluated TST systems. The proposed CMW-ATC framework obtains the lowest SER among the sequence-level and semi-supervised methods on all three evaluation sets. Relative to the labelled-data CTC baseline, CMW-ATC reduces SER by 24.0–28.7%, while reducing SER relative to ATC-R by 15.9–20.6%. Teacher–Student improves over the labelled-data CTC baseline, while ATC-R provides a further reduction in SER.

Compared with the other TST approaches, CMW-ATC obtains lower SER than the clustering baseline on all three evaluation sets and lower SER than the onset-based S&C method on HMR, while requiring only sequence-level stroke annotations. A key methodological distinction is that CMW-ATC performs sequence-level transcription with CTC and uses learned stroke-transition information to weight alternative stroke candidates at uncertain positions. In contrast, the clustering method first groups acoustic segments and maps the resulting clusters to stroke labels, while S&C classifies detected strokes from local acoustic segments. OTM achieves a lower SER than CMW-ATC on HMR, but requires onset-level annotations for training. Thus, CMW-ATC achieves the lowest SER among the evaluated sequence-level and clustering-based methods without requiring frame- or onset-level stroke annotations.

HMR is the most challenging evaluation setting, with recordings from 45 artists and vocal and instrumental accompaniment across four *tālas*. The higher SER observed on HMR may be associated with this greater acoustic and performer variability; nevertheless, CMW-ATC maintains the lowest SER among the evaluated sequence-level and semi-supervised methods.

TABLE I
DATASET STATISTICS

Dataset	Train (min)	Val. (min)	Test (min)
TID+TSD	103+412*	24	24
HMR	212	60	30
TID-S	81+324*	20	20

* indicates augmented data.

TABLE II
STROKE ERROR RATE (SER, %) ACROSS TST SYSTEMS.

Data	Strongly Supervised		Clustering	Sequence-level / Semi-Supervised			
	OTM [4]	S&C [1]		Based [7]	CTC Teacher-Student	ATC-R [14]	Proposed CMW-ATC
TID+TSD	—	—	48.2	25.1	24.2 (↓3.6%)	22.9 (↓8.8%)	18.3 (↓27.1%)
HMR	28.5	67.4	84.1	43.8	42.7 (↓2.5%)	39.6 (↓9.6%)	33.3 (↓24.0%)
TID-S	—	—	42.3	17.8	16.7 (↓6.2%)	16.0 (↓10.1%)	12.7 (↓28.7%)

Parenthesized values denote relative SER reduction (↓) relative to the CTC (Labelled Data) baseline. “—” denotes not applicable, since the required onset-level annotations for training the corresponding baselines are unavailable for the dataset.

TABLE III
ABLATION STUDY OF CONFIDENCE ESTIMATION AND CANDIDATE WEIGHTING IN THE PROPOSED CMW-ATC FRAMEWORK.

Method	Confidence	Candidate formulation / weighting	TID+TSD	HMR	TID-S
ATC-R [14]	CTC-based	Wildcard *	22.9	39.6	16.0
A1	S-CEM	Wildcard *	21.4	37.8	14.8
A2	S-CEM	All strokes – Uniform weighting	21.2	38.1	14.6
A3	S-CEM	All strokes – Forward Markov weighting	20.1	35.9	13.8
A4	CTC-based	All strokes – CMW weighting	20.7	36.9	14.2
Proposed CMW-ATC	S-CEM	All strokes – CMW weighting	18.3	33.3	12.7

Table III summarizes the ablation studies on confidence estimation and candidate weighting. Under the wildcard formulation, replacing CTC-based token confidence with S-CEM (ATC-R versus A1) reduces SER by 4.5–7.5% relative across the three evaluation sets. A4 and CMW-ATC provide a controlled comparison because they use the same set of all-stroke candidates and the same CMW weighting. Replacing CTC-based confidence with S-CEM in this setting reduces SER by 9.8–11.6%, showing the contribution of the proposed confidence estimator.

The effect of candidate weighting is evaluated through A2, A3, and the proposed CMW-ATC, which use the same S-CEM confidence and all-stroke candidates. Replacing uniform weighting in A2 with forward Markov weighting in A3 reduces SER by 5.2–5.8% relative. Adding the reliable right-boundary transition in CMW-ATC further reduces SER by 7.2–9.0% relative to A3. These results show that both Markov transition weighting and right-boundary information improve performance.

Fig. 1 shows the confidence distributions of correct and incorrect teacher-predicted strokes on TID+TSD. With CTC posterior confidence, the distributions of correct and incorrect predictions overlap considerably. In contrast, S-CEM gives higher confidence mainly to correct predictions and lower confidence to incorrect predictions, producing a clearer separation between the two distributions.

A. Transcription Error Analysis

To further analyze the transcription results, we decompose the errors into substitutions (S), deletions (D), and

insertions (I). Table IV shows that substitutions remain the largest error category for CMW-ATC across all three evaluation sets, while all three error types are reduced compared with the CTC baseline. This indicates that the proposed approach improves transcription performance across different types of sequence errors, with substitution errors remaining the primary source of residual errors.

The largest relative reduction is observed for insertion errors, with reductions of 38.9%, 41.5%, and 44.4% on TID+TSD, HMR, and TID-S, respectively. The candidate construction, however, preserves the number of pseudo-label positions and does not explicitly add or remove stroke positions. The reductions in insertion and deletion errors should therefore be viewed as effects of the resulting student model rather than as direct corrections performed by the candidate graph. CMW-ATC instead focuses on handling uncertainty in stroke identity at low-confidence positions, which contributes directly to the reduction in substitution errors. Nevertheless, substitution errors remain the dominant error category across all three evaluation sets.

VI. CONCLUSION

We presented a semi-supervised framework for Stroke Transcription that combines stroke-level confidence estimation with Confidence-Guided Markov Weighting Alternative Temporal Classification (CMW-ATC). S-CEM identifies uncertain teacher-predicted strokes, while CMW-ATC represents the corresponding uncertain spans using alternative candidate sequences and weights them using learned stroke-transition probabilities and reliable boundary information.

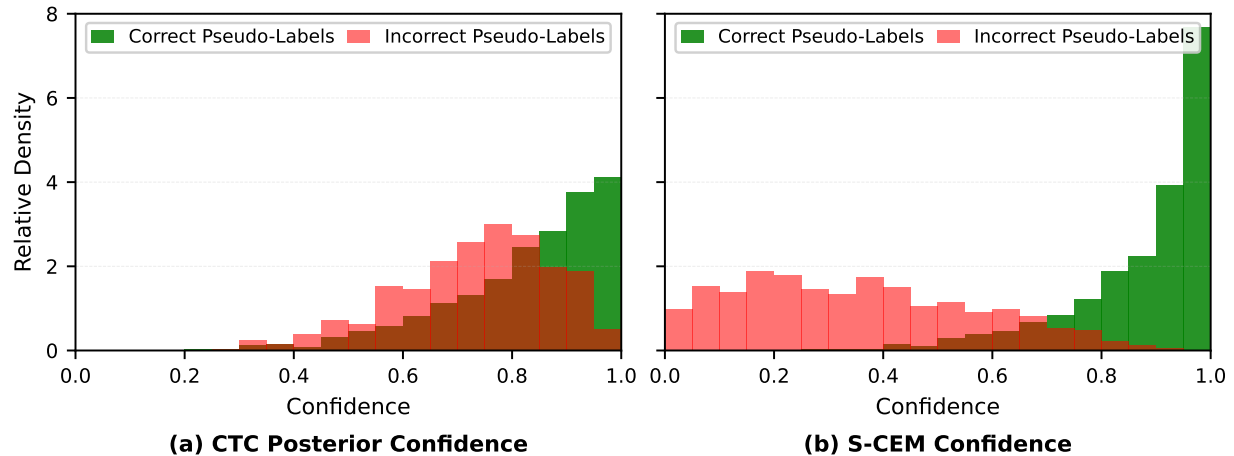


Fig. 1. Confidence distributions of correct and incorrect teacher-predicted strokes on the TID+TSD test set.

TABLE IV
OVERALL ERROR DISTRIBUTION FOR TID+TSD, HMR, AND
TID-S. VALUES IN PARENTHESES DENOTE THE RELATIVE
CONTRIBUTION OF EACH ERROR TYPE TO SER.

Dataset	Method	SER	S	D	I
TID+TSD	CTC	25.1	15.2 (60.6%)	4.5 (17.9%)	5.4 (21.5%)
	CMW-ATC	18.3	11.5 (62.9%)	3.4 (18.8%)	3.3 (18.3%)
HMR	CTC	43.8	22.4 (51.1%)	12.0 (27.4%)	9.4 (21.5%)
	CMW-ATC	33.3	18.9 (56.7%)	9.0 (26.9%)	5.5 (16.4%)
TID-S	CTC	17.8	10.8 (60.7%)	3.4 (19.1%)	3.6 (20.2%)
	CMW-ATC	12.7	8.3 (65.7%)	2.4 (18.7%)	2.0 (15.7%)

The proposed system obtains SERs of 18.3, 33.3, and 12.7 on TID+TSD, HMR, and TID-S, respectively, corresponding to relative SER reductions of 24.0–28.7% over the labelled-data CTC baseline and 15.9–20.6% over ATC-R. The ablation studies show contributions from S-CEM, forward Markov weighting, and the additional use of reliable right-boundary information in the complete CMW formulation. Error analysis shows reductions in substitution, deletion, and insertion errors, although CMW-ATC directly handles uncertainty in stroke identity and does not explicitly model insertion or deletion errors. Future work will investigate broader low-resource TST settings and further reductions in annotation requirements.

REFERENCES

- [1] P. Rao *et al.*, “Automatic stroke classification of tabla accompaniment in hindustani vocal concert audio,” *arXiv preprint arXiv:2104.09064*, 2021.
- [2] R. Gowriprasad and K. S. R. Murty, “Onset detection of tabla strokes using lp analysis,” in *2020 International Conference on Signal Processing and Communications (SPCOM)*. IEEE, 2020, pp. 1–5.
- [3] R. MA, A. Bhattacharjee, and P. Rao, “Four-way classification of tabla strokes with models adapted from automatic drum transcription,” in *ISMIR*, 2021.
- [4] R. M. Ananthanarayana, A. Bhattacharjee, and P. Rao, “Four-way classification of tabla strokes with transfer learning using western drums,” *Transactions of the International Society for Music Information Retrieval*, vol. 6, no. 1, 2023.
- [5] R. B. Kodag and V. Arora, “Meta-learning-based percussion transcription and *tāla* identification from low-resource audio,” *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- [6] R. Kodag and V. Arora, “Weakly supervised tabla stroke transcription via an adaptive dynamic rhythm language model (adrm),” *arXiv preprint arXiv:2608.30314*, 2026.
- [7] R. Gowriprasad, J. Kuriakose, R. Aravind, H. A. Murthy *et al.*, “Unsupervised stroke clustering for developing common stroke set for percussion in indian art music,” *The Journal of the Acoustical Society of America*, vol. 158, no. 5, pp. 3711–3723, 2025.
- [8] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, “Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks,” in *ICML*, 2006, pp. 369–376.
- [9] V. Pratap, A. Hannun, G. Synnaeve, and R. Collobert, “Star temporal classification: Sequence classification with partially labeled data,” *arXiv preprint arXiv:2201.12208*, 2022.
- [10] X. Cai, J. Yuan, Y. Bian, G. Xun, J. Huang, and K. Church, “W-ctc: a connectionist temporal classification loss with wild cards,” in *International Conference on Learning Representations*, 2022.
- [11] D. Gao, M. Wiesner, H. Xu, L. P. Garcia, D. Povey, and S. Khudanpur, “Bypass temporal classification: Weakly supervised automatic speech recognition with imperfect transcripts,” *arXiv preprint arXiv:2306.01031*, 2023.
- [12] D. Gao, H. Xu, D. Raj, L. P. G. Perera, D. Povey, and S. Khudanpur, “Learning from flawed data: Weakly supervised automatic speech recognition,” in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2023, pp. 1–8.
- [13] N. Moritz, T. Hori, and J. Le Roux, “Semi-supervised speech recognition via graph-based temporal classification,” in *ICASSP*. IEEE, 2021, pp. 6548–6552.
- [14] H. Zhu, D. Gao, G. Cheng, D. Povey, P. Zhang, and Y. Yan, “Alternative pseudo-labeling for semi-supervised automatic speech recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 3320–3330, 2023.
- [15] S. Asmussen and A. Hobolth, “Markov bridges, bisection and variance reduction,” in *Monte Carlo and Quasi-Monte Carlo Methods 2010*. Springer, 2012, pp. 3–22.

- [16] F. Baltazar-Larios and L. J. R. Esparza, “A novel method and comparison of methods for constructing markov bridges: F. baltazar-larios, ljr esparza,” *Computational Statistics*, vol. 40, no. 9, pp. 4969–4990, 2025.
- [17] N. Ravi, R. T. Chaganti, V. Arora *et al.*, “Asr confidence estimation using true class lexical similarity score,” in *Proc. Interspeech 2025*, 2025, pp. 3658–3662.
- [18] P. Chordia, “Segmentation and recognition of tabla strokes,” in *ISMIR*, vol. 20056, 2005, pp. 107–114.
- [19] K. Narang and P. Rao, “Acoustic features for determining goodness of tabla strokes,” in *ISMIR*, 2017, pp. 257–263.
- [20] S. Gupta, A. Srinivasamurthy, M. Kumar, H. A. Murthy, and X. Serra, “Discovery of syllabic percussion patterns in tabla solo recordings,” in *ISMIR*, 2015.
- [21] A. Anantapadmanabhan, A. Bellur, and H. A. Murthy, “Modal analysis and transcription of strokes of the mridangam using non-negative matrix factorization,” in *ICASSP*. IEEE, 2013, pp. 181–185.
- [22] A. Anantapadmanabhan, J. Bello, R. Krishnan, and H. Murthy, “Tonic-independent stroke transcription of the mridangam,” in *Audio engineering society conference: 53rd international conference: Semantic audio*. Audio Engineering Society, 2014.
- [23] J. Kuriakose, J. C. Kumar, P. Sarala, H. A. Murthy, and U. K. Sivaraman, “Akshara transcription of mridangam strokes in carnic music,” in *Twenty First National Conference on Communications (NCC)*. IEEE, 2015, pp. 1–6.
- [24] W. He, Z. Jiang, T. Xiao, Z. Xu, and Y. Li, “A survey on uncertainty quantification methods for deep learning,” *ACM Computing Surveys*, vol. 58, no. 7, pp. 1–35, 2026.
- [25] C. Corbière, N. Thome, A. Bar-Hen, M. Cord, and P. Pérez, “Addressing failure prediction by learning model confidence,” *Advances in neural information processing systems*, vol. 32, 2019.
- [26] Q. Li, D. Qiu, Y. Zhang, B. Li, Y. He, P. C. Woodland, L. Cao, and T. Strohmman, “Confidence estimation for attention-based sequence-to-sequence models for speech recognition,” in *ICASSP*. IEEE, 2021, pp. 6388–6392.
- [27] N. Ravi, V. Arora *et al.*, “Teles: Temporal lexeme similarity score to estimate confidence in end-to-end asr,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 4399–4408, 2024.
- [28] A. Srinivasamurthy, A. Holzapfel, A. T. Cemgil, and X. Serra, “A generalized bayesian model for tracking long metrical cycles in acoustic music signals,” in *ICASSP*. IEEE, 2016, pp. 76–80.
- [29] M. Yarmohammadi, “Semi-orthogonal low-rank matrix factorization for deep neural networks,” *Interspeech*, 2018.
- [30] J. V. Psutka, J. Švec, and A. Pražák, “Cnn-tdnn-based architecture for speech recognition using grapheme models in bilingual czech-slovak task,” in *International Conference on Text, Speech, and Dialogue*. Springer, 2021, pp. 523–533.